FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIV

JEE (Main)-2022

TEST DATE: 09-06-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A uniformly charged non conducting sphere of charge Q and radius R is kept in a uniform electric field E. A very small part of area dA is removed from the sphere and sphere is released from rest when position vector of hole with respect to center of sphere is perpendicular to electric field. Which of the following is correct?

(A) Initial torque on sphere about its CM is $\frac{QERdA}{4\pi R^2}$

(B) Maximum angular velocity of sphere is $\sqrt{\frac{3QEdA}{4\pi mR^3}}$

(C) Maximum angular velocity of sphere is $\sqrt{\frac{5QEdA}{4\pi mR^3}}$

(D) Sphere will not rotate.

2. Angle of declination at any point on the equator of earth is $\phi \neq 0$

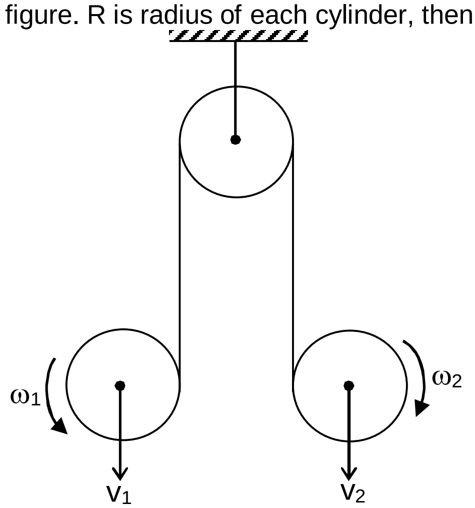
(A) ϕ will increase if observer will move along a longitude towards north pole

(B) ϕ will remain same if observer will move along a longitude toward north pole.

(C) ϕ must increase if observer moves along equator towards east.

(D) ϕ will remain constant if observer moves along equator towards east.

3. Ends of an inextensible string is wrapped tightly over two cylinders & string is allowed to pass over a fixed pulley as shown. Linear & angular velocities of cylinder at any instant are indicated in figure. R is radius of each cylinder, then



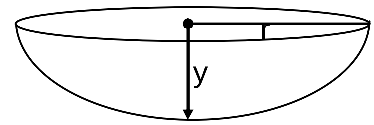
(A) $v_1 = \omega_1 R$

(B) $v_2 = \omega_2 R$

(C) $v_1 + v_2 = (\omega_1 + \omega_2)R$

(D) $v_1 - v_2 = (\omega_1 - \omega_2)R$

4. A thin film of a liquid having mass m is formed on a ring radius r. Thickness of film is equal to thickness of ring and angle of contact is zero. When ring is kept horizontal in air then center of film goes down by distance y $(y << r)$. Find surface tension of film.



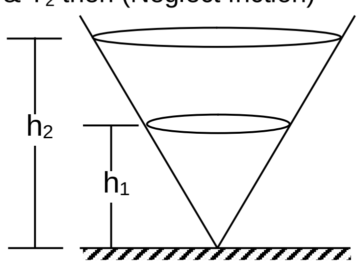
(A) $\frac{mg}{2\pi y}$

(B) $\frac{mg}{4\pi y}$

(C) $\frac{2mg}{\pi y}$

(D) $\frac{4mg}{\pi y}$

5. Two particles are moving along horizontal circular path on inner surface of right circular cone. Height of circular paths are h_1 and h_2 as shown and corresponding time period of particles are T_1 & T_2 then (Neglect friction)



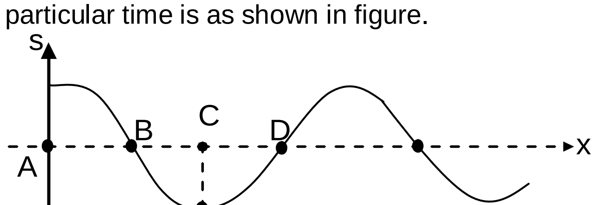
(A) $\frac{T_1}{T_2} = \frac{h_1}{h_2}$

(B) $\frac{T_1}{T_2} = \frac{h_2}{h_1}$

(C) $\frac{T_1^2}{T_2^2} = \frac{h_1}{h_2}$

(D) $\frac{T_1^2}{T_2^2} = \frac{h_2}{h_1}$

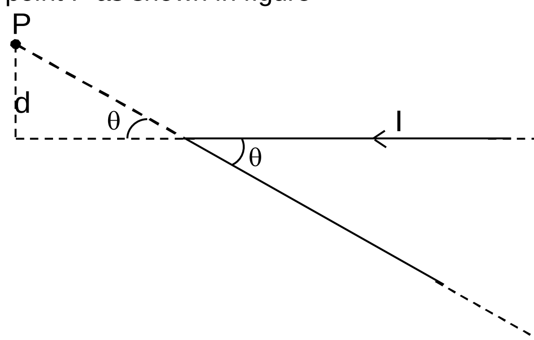
6. A plane sound wave is traveling along positive x-axis. Its displacement versus position at particular time is as shown in figure.



Which of the following is correct?

(A) Air at point A has velocity in positive direction and is a rarefaction.
(B) Air at point B has velocity in positive direction and is a rarefaction.
(C) Air at point C has zero velocity and is a compression.
(D) Air at point D has velocity in negative direction and is a rarefaction.

7. A long straight wire carrying current I is bent at mid point to form angle θ. Find magnetic field at point P as shown in figure



(A) $\frac{\mu_0 i}{4\pi d}(1+\cos\theta)$

(B) $\frac{\mu_0 i}{4\pi d}(1-\sin\theta)$

(C) $\frac{\mu_0 i}{4\pi d}(1-\cos\theta)$

(D) $\frac{\mu_0 i}{4\pi d}(1+\sin\theta)$

8. If rms speed of molecules of a monatomic gas is increased 1.5 times through adiabatic process then what is ratio of initial volume to final volume?

(A) $\left(\frac{2}{3}\right)^{5/4}$

(B) $\left(\frac{3}{2}\right)^{3/4}$

(C) $\left(\frac{3}{2}\right)^{3}$

(D) $\left(\frac{2}{3}\right)^{2}$

9. A ring of radius 3 m is uniformly charged. There are two points on the axis and on the same side of ring where ratio of electric potential and electric field intensity is 10 m. What is the separation between these two points?

(A) 2m
(B) 4m
(C) 6m
(D) 8m

10. A capillary tube is immersed vertically in water and the height of the water column is h_1. If arrangement is taken into a mine at depth d, the height of water column is h_2. If R is radius of earth than $\frac{h_1}{h_2}$ is

(A) $\left(1-\frac{d}{R}\right)$

(B) $\left(1+\frac{d}{R}\right)$

(C) $\frac{R-d}{R+d}$

(D) $\frac{R+d}{R+d}$

11. Voltage drop across p side, n side and depletion region of pn junction in forward bias are V_p, V_n and V_d respectively then,

(A) $V_d > V_p$

(B) $V_d < V_n$

(C) $V_d = V_p$

(D) $V_d = V_n$

12. If pressure (p), density (d) and velocity (v) are taken as base quantities in place of mass, length and times then dimension of plank's constant is

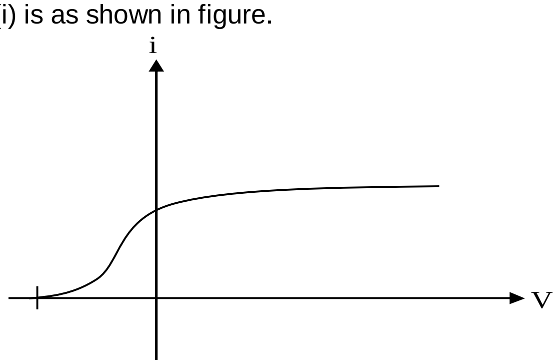
(A) pd^2v

(B) p^2dv

(C) $pd^{-2}v^{-1}$

(D) p, d & v cannot be taken as base quantities

13. In photoelectric experiment, graph obtained for collector plate potential (V) versus photocurrent (i) is as shown in figure.



Which of the following can be concluded from the graph?

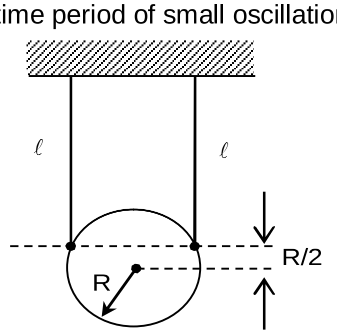
(A) Photocurrent is independent of intensity of incident light.

(B) All the photoelectrons do not have same kinetic energy

(C) Photoelectron emission is instantaneous

(D) Stopping potential depends on frequency of incident light.

14. A disc of mass m and radius R is attached to ceiling with the help of ropes of length *l*. Find the time period of small oscillation of disc in the plane of disc.



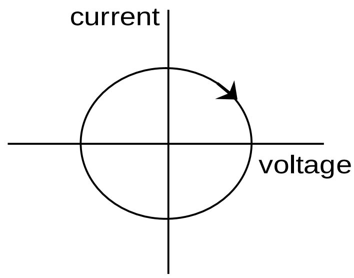
(A) $2\pi\sqrt{\frac{l+R/2}{g}}$

(B) $2\pi\sqrt{\frac{l^2+(R/2)^2}{g(R/2+l)}}$

(C) $2\pi\sqrt{\frac{l}{g}}$

(D) None of these

15. The graph shows the current versus the voltage in a driven RLC circuit at a fixed frequency. The arrow indicates the direction that this curve is drawn as time progresses.



In this plot, the
(A) Current lags the voltage by 90 degrees
(B) Current leads the voltage by 90 degrees
(C) Current and voltage are in phase
(D) Current and voltage are 180 degrees out of phase

16. A pin 10 mm tall is used as an object in front of a mirror and the image formed is erect and 2mm tall.
(A) Moving the pin closer to the mirror will make the image larger
(B) The image must be real
(C) The mirror must be converging
(D) Both (B) and (C) are correct

17. Ionization energy of a hydrogen-like ion *A* is greater than that of another hydrogen-like ion *B*. Let *r*, U, *E* and *L* represent the radius of the orbit, speed of the electron, energy of the atom and orbital angular momentum of the electron respectively. In ground state:
(A) $r_A > r_B$
(B) $U_A > U_B$
(C) $E_A > E_B$
(D) $L_A > L_B$

18. A shunt of resistance 1 Ω is connected across a galvanometer of 9Ω resistance. A current of 5 A gives full scale deflection in the galvanometer. The current that will give full scale deflection in the absence of the shunt is:
(A) 5.5 A
(B) 0.5 A
(C) 0.004 A
(D) 0.045 A

19. A monochromatic light is used in Young's double slit experiment. When one of the slits is covered by a transparent sheet of thickness 1.8 mm and made of material of refractive index μ_1, then number of fringes which crosses a point on the screen is 18. when another sheet of thickness 3.6 mm and made of material of refractive index μ_2 is used, number of fringes which shift is 9. Relation between μ_1 and μ_2 is given by

(A) $4\mu_2 - \mu_1 = 3$

(B) $4\mu_1 - \mu_2 = 3$

(C) $3\mu_2 - \mu_1 = 4$

(D) $2\mu_1 - \mu_2 = 4$

20. The binding energies of the nuclei of ${}_2^4He$, ${}_3^7Li$, ${}_6^{12}C$ & ${}_7^{14}N$ are 28, 52, 90, 98 MeV respectively. Which of these is most stable?

(A) ${}_2^4He$

(B) ${}_3^7Li$

(C) ${}_6^{12}C$

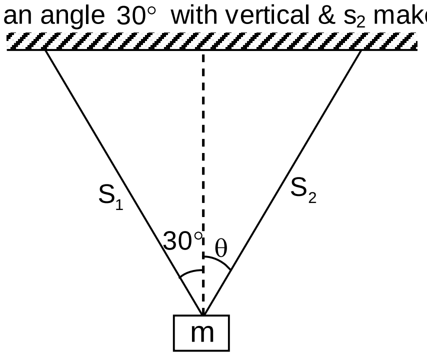
(D) ${}_7^{14}N$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

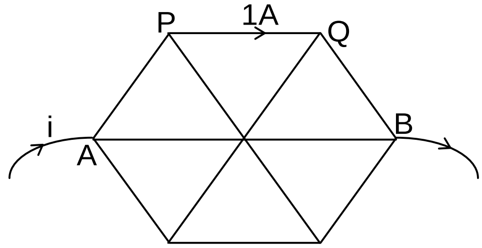
21. Two particles are projected vertically up with the same speed 40 m/s on the surface of planets where accelerations due to gravity are 5 m/s^2 and 10 m/s^2 respectively. What is the ratio of distance covered in 6 sec?

22. A block of mass m is hanging in equilibrium with the help of two strings s_1 & s_2. String s_1 makes an angle $30°$ with vertical & s_2 makes an angle θ with vertical as shown.



Find θ in degree to have minimum tension in string S_2.

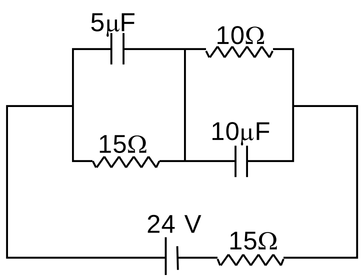
23. 12 identical wire is used in a network as shown. A current i enters into network at A and leaves at B such that current in wire PQ becomes 1A. Find the value of i in ampere.



24. A rough incline plane is inclined at 30° and its length is 5m. A block of mass 4 kg is pulled upward to top from bottom. The man pulling the block needs to supply 400 J for this while applying force parallel to incline plane. What is work done by the man for same process if incline plane has inclination 60°. (g = 10 m/s^2)

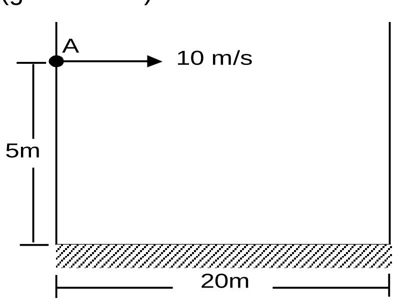
25. Find percentage error in measurement of moment of inertia of an annular disc if measured value of inner radius is (3.0 ± 0.1) cm, outer radius is (4.0 ± 0.1) cm and mass is (50 ± 1)g.

26. What is total energy (in μJ) stored in the capacitors of given circuit in steady state?

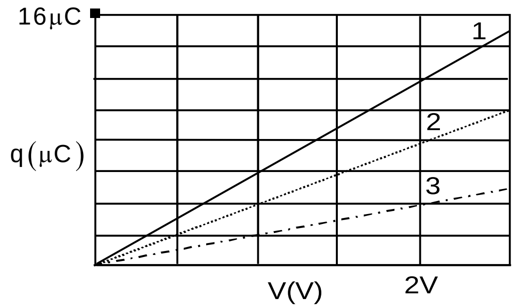


27. A hollow cylinder and a block are released on a rough inclined plane. Cylinder rolls down without slipping and incline plane makes angle of 45^o with horizontal. Find coefficient of friction between block and inclined plane if both cylinder and block move with same acceleration.

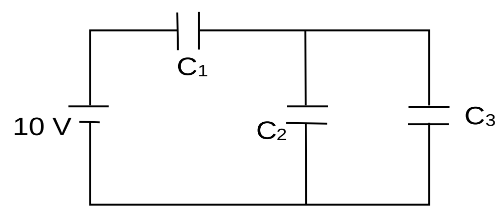
28. A particle is projected horizontally from a point A between two parallel walls as shown. All collisions are elastic and neglect friction. Find time period (in second) of motion of particle. (g = 10 m/s^2)



29. Plot 1 in figure gives the charge q that can be stored on capacitor 1 versus the electric potential V set up across it. Plots 2 and 3 are similar plots for capacitor 2 and 3, respectively. Now these capacitors are connected as in circuit (b). What is the charge (in μC) stored on capacitor 2 in that circuit?

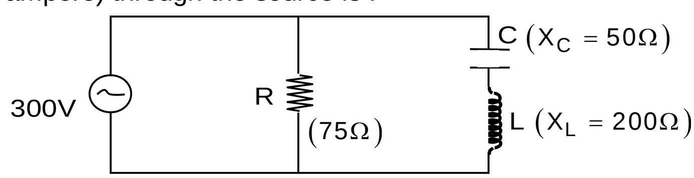


(a)



(b)

30. In the circuit diagram shown, X_C = 50 ohm, X_L = 200 ohm & R = 75 ohm. The effective current (in ampere) through the source is :



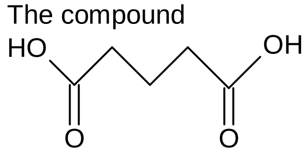
Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The compound



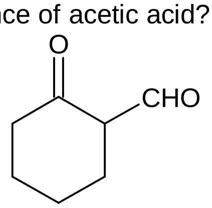
is subjected to following treatment stepwise:

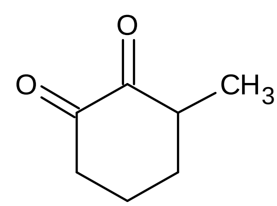
(i) $SOCl_2$ (ii) NH_3 (iii) $LiAlH_4/H_3O^+$

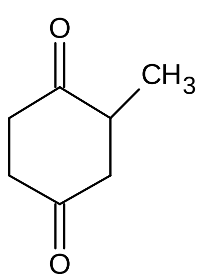
The final product of the reaction is

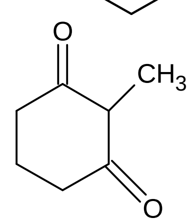
(A) Propanamide
(B) 5-Amino pentane-1-ol
(C) 1, 5-Propanediamine
(D) 1, 5-Pentanediamine

32. Which is the product formed when 2-methylcyclohexanone is treated with selenium dioxide in presence of acetic acid?

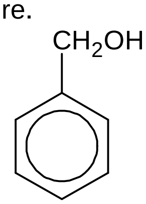
(A)


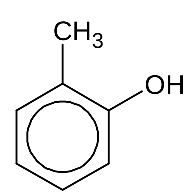
(B)


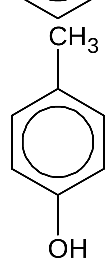
(C)


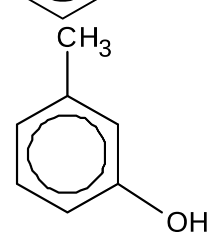
(D)


33. A crystalline solid having molecular formula C_7H_8O reacts with sodium to liberate hydrogen. It is insoluble in water but soluble in dilute aqueous NaOH upon treatment with Br_2 in CCl_4, Only one monobrominated compound with the formula C_7H_7OBr is formed. The original compound has the structure.

(A)


(B)


(C)


(D)


34. Which of the following statement is incorrect about ores?
(A) Calamine and siderite are carbonate.
(B) Argentite and cuprite are oxide.
(C) Zinc blende and iron pyrite are sulphides
(D) Malachite and azurite are ores of copper.

35. A white solid (A) dissolves in water. The aqueous solution gives a white precipitate (B) when NaOH or NH_4OH is added. (B) dissolve in excess of NaOH but not in NH_4OH. $BaCl_2$ solution, when added to a solution of (A) gives a white precipitate (C) which is insoluble in dilute HCl. (A) may be
(A) $Al_2(SO_4)_3$
(B) $ZnSO_4$
(C) $SnSO_4$
(D) All of the above

36. Solution of salt (X) gives a scarlet coloured precipitate with KI solution which dissolves in its excess. The colorless solution with ammonium chloride in presence of alkali forms red brown precipitate. The compound (X) is
(A) Mohr's salt
(B) Glauber's salt
(C) Epsom salt
(D) Corrosive sublimate

37. The product obtained when $MnSO_4$ solution is boiled with PbO_2 and concentrated HNO_3 is
(A) MnO_2
(B) $HMnO_4$
(C) Mn_3O_4
(D) $PbMnO_4$

38. $K_4\left[Fe(CN)_6\right]$ can be used to detect which of the following ions out of the following $Fe^{+2}, Fe^{+3}, Zn^{+2}, Cu^{+2}$ and Cd^{+2} ?
(A) Fe^{+2}, Fe^{+3}
(B) $Fe^{+3}, Zn^{+2}, Cu^{+2}$
(C) all but not Fe^{+3}
(D) all but not Fe^{+2}

39. A graph is plotted between E_{cell} of quinhydrone half cell and pH of solution. Intercept is 0.699 V. At what pH, E_{cell} will become 0.492V?
(A) 3.5
(B) 4.2
(C) 7
(D) 10

40. In which of the following case would the probability of finding an electron residing in a d_{xy} orbital be zero?
(A) XY and YZ plane
(B) XY and XZ plane
(C) XZ and YZ plane
(D) Z direction, YZ and XZ plane

41. One litre of an ideal gas diffuse in 5 minutes at S.T.P, How much time it would take for the same diffusion at 1 atm and $273°C$?
(A) 7.1 min
(B) 10 min
(C) 14.1 min
(D) cannot say

42. 0.85% aqueous solution of $NaNO_3$ is apparently 90% dissociated. The osmotic pressure of solution at 300 K is
(A) 4.67 atm
(B) 46.74 atm
(C) 2.46 atm
(D) 4.67 mm of Hg

43. Calcium crystallizes in a face centered cubic unit cell with edge length (a) = 0.556 nm. Suppose this sample of Ca contained 0.1% Frenkel defect in the crystal. The density of crystal
(A) change by 0.05%
(B) change by 0.1%
(C) change by 0.5%
(D) does not change

44. A base, $C_{17}H_{21}O_4N$ is soluble in water to the extent of 1.7 gm/lit and a saturated solution has a pH of 10.08. The value of the ionization constant, K_b for this base is: $\left[\text{Antilog}\left(-3.92\right)=1.2\times10^{-4}\right]$
(A) 5.6×10^{-6}
(B) 1.2×10^{-6}
(C) 2.62×10^{-6}
(D) 9.04×10^{-6}

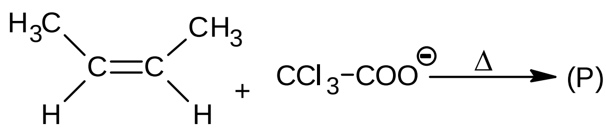
45. For the elementary reaction $2NO+H_2\longrightarrow N_2O+H_2O$; the half life time is 19.2 sec at $820°C$ when partial pressure of NO and H_2 are 600 mm of Hg and 10mm of Hg respectively. The value of half life time at the same temperature when partial pressure of NO and H_2 are 600 mm Hg and 20 mm Hg respectively would be:
(A) 19.2 sec
(B) 10 sec
(C) 830 sec
(D) 415 sec

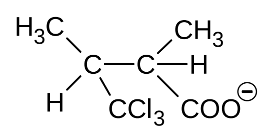
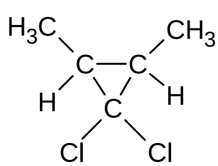
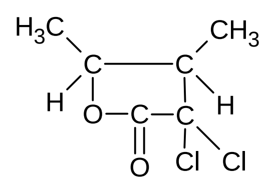
46. The weight of acetic anhydride that must be added to 75 gm of 92% acetic acid to make the concentration of acetic acid 100% would be
(A) 22 gm
(B) 38 gm
(C) 34 gm
(D) 56 gm

47. Solubility of a substance which dissolve with a decrease in volume and absorption of heat will be formed by
(A) high P and high T
(B) low P and low T
(C) high P and low T
(D) low P and high T

48. A reversible isothermal evaporation of 90gm of water is carried out at $100°C$. Heat of evaporation of water is 9.72 KCal/mol. Assuming water vapour to behave like an ideal gas. What is the change in internal energy of the system?
(A) 48.6 KCal
(B) 52.33 KCal
(C) 44.87 KCal
(D) 56.06 KCal

49. Langmuir adsorption isotherm is applicable to
(A) Physical adsorption
(B) Chemical adsorption
(C) Any monolayer adsorption
(D) Both physical adsorption and activated adsorption

50. 

Product (P) of reaction is
(A) No reaction occurred
(B) 
(C) 
(D) 

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. How many number of edges present in a truncated tetrahedron?

52. In a crystal of XY, X form CCP lattice and Y are present at all tetrahedral voids. If particles along one of the body diagonal are removed, then formula of compound is X_nY_m. The value of (n+m) is?

53. Calculate the percentage of dissociation of an electrolyte XY_2 (normal molar mass = 164) in water. If the observed molar mass by measuring elevation in boiling point is 65.6?

54. A saturated solution in $PQ_{(s)}$ and $PR_{(s)}$ has conductivity of $290 \times 10^{-10}\,Scm^{-1}$. If K_{sp} of $PQ_{(s)}$ and $PR_{(s)}$ are 3×10^{-14} and 1×10^{-14} respectively. The limiting molar conductivity of P^+, Q^-, R^- are 50 Scm^2mol^{-1}, X Scm^2mol^{-1} and 20 Scm^2mol^{-1} respectively. The value of X is

55. The sum of number of hexagonal faces and cubical faces present in a truncated octahedron is

56. The sum of mono and dihalo substituted isomers are possible for benzene and borazole (Inorganic benzene) is

57. The rate of first order reaction is 0.04 $molL^{-1}s^{-1}$ at 10 seconds and 0.03 $molL^{-1}s^{-1}$ at 20 seconds after initiation of reaction. The half life period of reaction is?

58. How many number of Faraday is required to convert 5 mol of nitrobenzene to aniline?

59. If mass of one atom is $3.32\times10^{-23}\,gm$, then calculate number of nucleons (neutrons and protons) present in 4 atoms of the element.

60. In the reaction $AB_{(g)} \rightleftharpoons A_{(g)} + B_{(g)}$ at $300°C$, K_p for dissociation equilibrium is $2.56\times10^{-2}\,atm.$ If the total pressure at equilibrium is 1 atm, then the percentage dissociation of AB is approximately

Mathematics PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The equation of sphere concentric with the sphere $x^2+y^2+z^2-4x-6y-8z-5=0$ and which passes through the point (0,10) is
(A) $x^2+y^2+z^2+4x+6y-8z-7=0$
(B) $x^2+y^2+z^2-4x-6y-8z=0$
(C) $x^2+y^2+z^2-4x-6y-8z+5=0$
(D) None of these

62. The angle between the line $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z-3}{2}$ and the plane $3x+6y-2z+5=0$ is
(A) $\cos^{-1}\left(\frac{4}{21}\right)$
(B) $\sin^{-1}\left(-\frac{4}{21}\right)$
(C) $\sin^{-1}\left(\frac{6}{21}\right)$
(D) $\sin^{-1}\left(\frac{4}{21}\right)$

63. Let N be any four digit number say $x_1\ x_2\ x_3\ x_4$. Then maximum value of $\frac{N}{x_1+x_2+x_3+x_4}$ is equal to
(A) 1000
(B) $\frac{1111}{4}$
(C) 800
(D) None of these

64. Let α, β and γ be roots of $f(x)=x^3+x^2-5x-1=0$. Then $[\alpha]+[\beta]+[\gamma]$ where $[.]$ denotes the greatest integer function, is equal to
(A) 1
(B) –2
(C) 4
(D) –3

65. If $\left|z-\frac{4}{z}\right|=2,$ then find the maximum value of $|z|$ is equal to
(A) 1
(B) $2+\sqrt{2}$
(C) $\sqrt{3}+1$
(D) $\sqrt{5}+1$

66. If $x_k \geq 0, k = 1,2,3,4,5$ and $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ and $x_1 + x_2 + x_3 = 5$ then the number of integral solution of $(x_1, x_2, x_3, x_4, x_5)$
(A) 84
(B) 168
(C) 286
(D) 336

67. $\int \frac{\operatorname{cosec}^2 x - 2005}{\cos^{2005} x} dx$ is equal to
(A) $\frac{\cot x}{(\cos x)^{2005}} + c$
(B) $\frac{\tan x}{(\cos x)^{2005}} + c$
(C) $\frac{-(\tan x)}{(\cos x)^{2005}} + c$
(D) $-\frac{\cot x}{(\cos x)^{2005}} + c$

68. Two number a, b are chosen from the set of integers 1, 2, 3, ….., 39, then probability that the equation 7a – 9b = 0 is satisfied, is
(A) $\frac{1}{247}$
(B) $\frac{2}{247}$
(C) $\frac{4}{741}$
(D) $\frac{5}{741}$

69. Let m be a positive integer and $\Delta_r = \begin{vmatrix} 2r-1 & {}^mC_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix}$ then the value of $\sum_{r=0}^{m} \Delta r$ is given by
(A) 0
(B) $m^2 - 1$
(C) 2^m
(D) $2^m \sin^2(2^m)$

70. $f(x)=\begin{cases} 3[x]-5\frac{|x|}{x}, & x\neq 0 \\ 2, & x=0 \end{cases}$. Then $\int_{-3/2}^{2} f(x)dx$ is equal to ($[.]$ denotes the greatest integer function)

(A) $-\frac{11}{2}$

(B) $-\frac{7}{2}$

(C) – 6

(D) $-\frac{17}{2}$

71. If $\theta=\pi/7$, then $\cos\theta+\cos 3\theta+\cos 5\theta$ is

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{8}$

(D) $\frac{1}{16}$

72. If $\tan^{-1}(x+h)=\tan^{-1}(x)+(h\sin y)(\sin y)-(h\sin y)^2.\frac{\sin 2y}{2}+(h\sin y)^3.\frac{\sin 3y}{3}+$ ……..

Where $x\in(0,1)$, $y\in(\pi/4,\pi/2)$, then

(A) $y=\tan^{-1}x$
(B) $y=\sin^{-1}x$
(C) $y=\cot^{-1}x$
(D) $y=\cos^{-1}x$

73. PQ is a vertical pole with end P on level ground. R is the mid-point of PQ. At a point A on the level ground, the portion QR subtends an angle α at A. If AP = nPQ, then $\alpha=$

(A) $\frac{n}{2n^2+1}$

(B) $\frac{n}{n^2-1}$

(C) $\frac{n}{n^2+1}$

(D) None of these

74. If $f(x)=\begin{vmatrix} x+a^2 & x^4+1 & 3 \\ x+b^2 & 2x^4+2 & 3 \\ x+c^2 & 3x^4+7 & 3 \end{vmatrix}$, (where $x\neq 0$) and f' (x) = 0, then a^2, b^2, c^2 are in

(A) Geometric progression
(B) Arithmetic Progression
(C) Harmonic Progression
(D) Arithmetic Geometric Progression

75. $\int \frac{\cos 2\theta}{(\sin\theta + \cos\theta)^2} d\theta$ is equal to

(A) $\frac{-1}{\sin\theta + \cos\theta} + c$

(B) $\log|\sin\theta + \cos\theta| + c$

(C) $\log|\sin\theta - \cos\theta| + c$

(D) $\log(\sin\theta + \cos\theta)^2 + c$

76. Given a function f(x) continuous $\forall x \in R$ such that $\lim_{x\to 0}\left[f(x) + \log\left(1 - \frac{1}{e^{f(x)}}\right) - \log(f(x))\right] = 0$, then f(0) is

(A) 0
(B) 1
(C) 2
(D) 3

77. Nine hundred distinct N –digit numbers are to be formed by using 6, 8 and 9 only. The smallest value of N for which this is possible is

(A) 6
(B) 7
(C) 8
(D) 9

78. y = x + r and y = -x + r where r takes all decimal digits. Then the number of squares in xy plane formed by these lines with diagonals of 2 units length are

(A) 81
(B) 100
(C) 64
(D) 49

79. If $\sin^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4}......\infty\right) + \cos^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4}.....\infty\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then $\cot^{-1} x$ equals to

(A) $-\frac{\pi}{4}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

80. The mean of two samples of sizes 200 and 300 were found to be 25 and 10 respectively. Their standard deviation were 3 and 4 respectively. The variance of combined sample so size 500 is

(A) 64
(B) 65.2
(C) 67.2
(D) 64.2

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. $\lim\limits_{h\to 0} \sqrt{3}\,\dfrac{2\left\{\sqrt{3\sin\left(\frac{\pi}{6}+h\right)-\cos\left(\frac{\pi}{6}+h\right)}\right\}}{\sqrt{3}h\left(\sqrt{3}\cosh - \sinh\right)}$ is equal to

82. If P is any pt on the hyperbola $\dfrac{(x-1)^2}{9}-\dfrac{(y+1)^2}{16}=1$ and S_1, S_2 are its foci, then $\left|S_1P - S_2P\right|$ is equal to?

83. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to $\dfrac{(k+1)\sqrt{k}}{2k}$, where k is prime number, find k

84. The equation of the base of an equilateral triangle is x + y – 2 = 0 and one vertex is (2, -1). Then the length of the side of the triangle is $\sqrt{\dfrac{k}{k+1}}$, find k?

85. If $\tan x - \tan^2 x = 1$, then the value of $\tan^4 x - 2\tan^3 x - \tan^2 x + 2\tan x + 1$ is?

86. Let P_i and P_i' be the feet of perpendiculars drawn from foci S, S' on a tangent T to an ellipse whose length of semi major axis is 20. If $\sum\limits_{i=1}^{10}(SP_i)(S'P_i') = 2560$, then the value of eccentricity is:

87. A chord AB whose equation is ax + by + c = 0 (where a, b, c are in A.P.) Cuts the parabola $y^2 + 4y - 8x - 4 = 0$ at points A and B. The angle between the tangents drawn to the parabola at points A and B is k^0. Find k.

88. The value of 'a' for which the equation $(a^2+4a+3)x^2 + (a^2-a-2)x + (a+1)a = 0$ has more than two roots is –k, then the value of k + 2020 is

89. Equation of the normal to the curve $x^2 = 4y$ which passes through the point (1, 2) is $x + y = k - 12$, then k =

90. Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x – axis. If $\tan\phi.\tan\theta = 2$ and $2\alpha^2 - \beta^2 = k$, then find k + 11.